Cardinal inequality is reflexive and transitive on cardinal numbers

The Univalent Foundations Program

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Editorial note. The counted statement is the author's unlabelled lem environment at pinned-original-source.tex lines 787-794, the lemma on cardinal inequality that opens the sequence of order-theoretic results of section 10.2; the proof environment opens at line 795, so the pairing is by adjacency.

Editorial note. Numbering: the lemma carries no label, so its number is part of its identity. The retained counter values (chapter 10, section 2, theorem offset 7) print it as 10.2.8, which is the position obtained by counting the theorem-class environments of section 10.2 against the dump-confirmed anchors 10.2.4 (card:semiring, source line 730) and 10.2.6 (card:exp, source line 759).

Editorial note. The proof proves transitivity in full and records reflexivity as immediate; the retained range is the whole author proof environment, and no part of the statement is left to the reader beyond that evident identity injection.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the pinned first-edition chapter, counted once: reflexivity and transitivity of cardinal inequality (unlabelled lemma, original Lemma 10.2.8).

Allowed standard prerequisites (editorial).

- Cardinal inequality. alpha<=beta means there is an injection from a representative of alpha to a representative of beta.. It is the relation whose preorder laws are asserted.
- Truncation induction on card. card is the 0-truncation of the type of sets, so a mere proposition indexed by card may be proved by assuming the index is cd{A}.. It is what lets the proof pass from cardinals to sets.
- Induction on propositional truncation. To prove a mere proposition from a propositional truncation it suffices to assume a witness.. It extracts the injections from the truncated inequalities.

Lemma 10.2.8. *Cardinal inequality is a preorder, i.e. for* $\alpha, \beta : \mathsf{Card}$ *we have*

$$\alpha \leq \alpha$$
$$(\alpha \leq \beta) \to (\beta \leq \gamma) \to (\alpha \leq \gamma)$$

Proof. As before, by induction on truncation. For instance, since $(\alpha \leq \beta) \to (\beta \leq \gamma) \to (\alpha \leq \gamma)$ is a mere proposition, by induction on 0-truncation we may assume α, β, and γ are $|A|_0$, $|B|_0$, and $|C|_0$ respectively. Now since $|A|_0 \leq |C|_0$ is a mere proposition, by induction on (-1)-truncation we may assume given injections $f : A \to B$ and $g : B \to C$. But then $g \circ f$ is an injection from A to C, so $|A|_0 \leq |C|_0$ holds. Reflexivity is even easier. $\square$